Log Analysis and Visualization with Splunk

Objective 3 - Log Analysis and Visualization with Splunk

Building on the manual and automated log analysis, this objective introduces Splunk, a Security Information and Event Management (SIEM) tool used in professional SOC environments.
In this stage, you'll learn how to ingest, query, and visualize Linux log data in Splunk
to identify suspicious activities at scale.

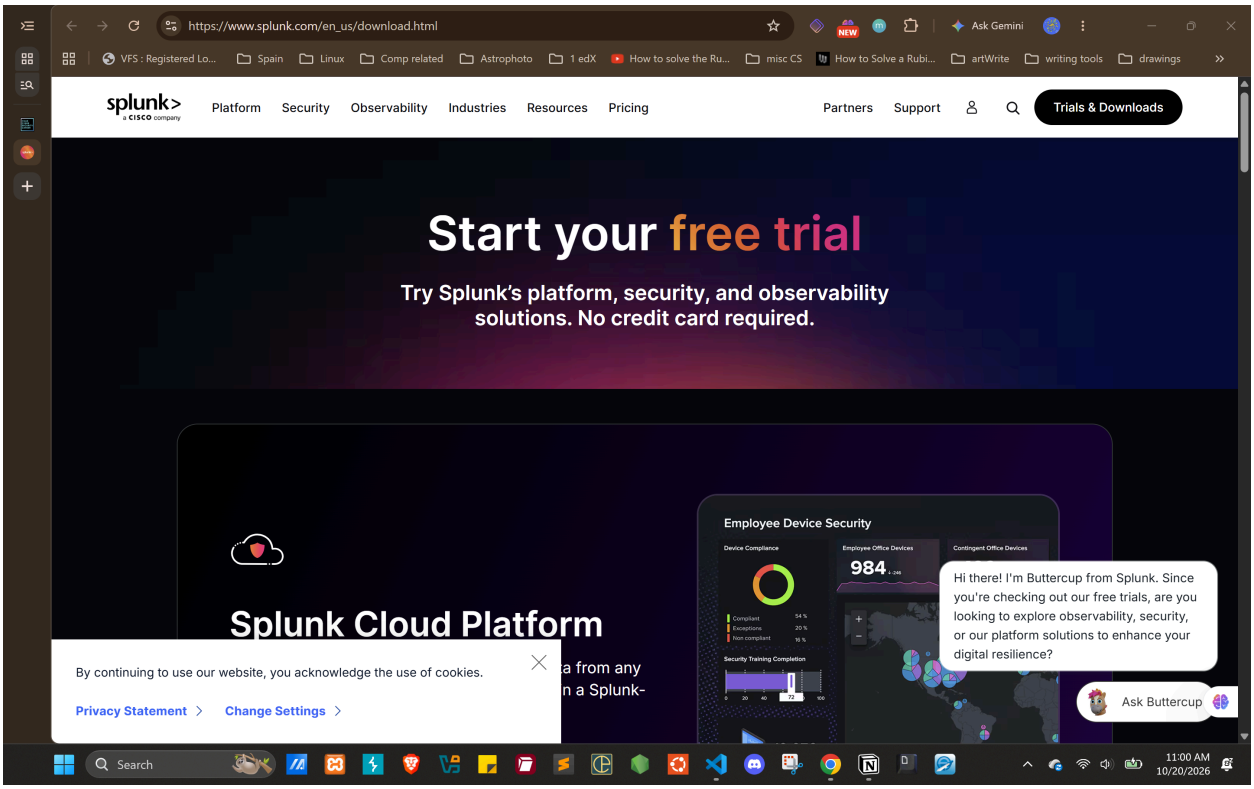
Required Tools / Prerequisites

1. Completion of Objectives 1 and 2 (familiarity with log structure and suspicious event indicators).
2. Access to a Splunk environment (Splunk Enterprise).
3. The Linux_2k.log file from LogHub.
4. Basic understanding of searching and filtering data in Splunk

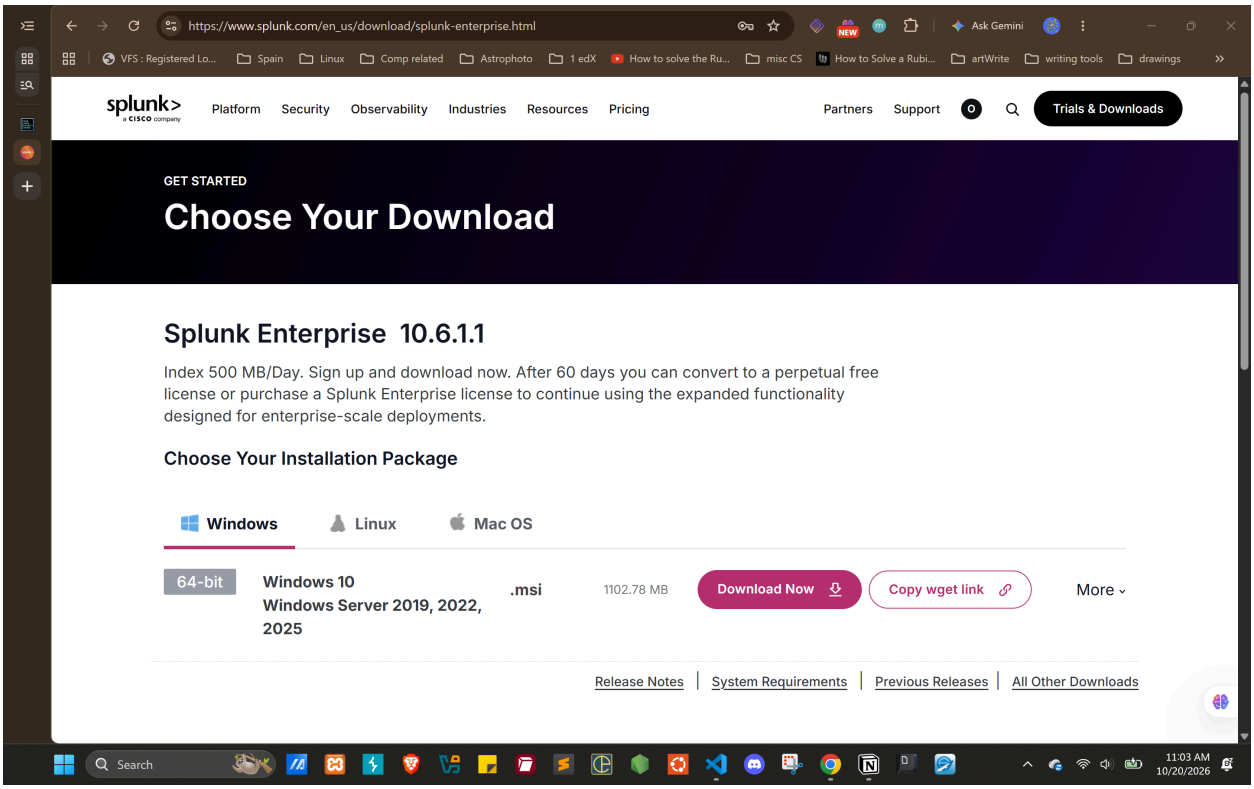
Instructions

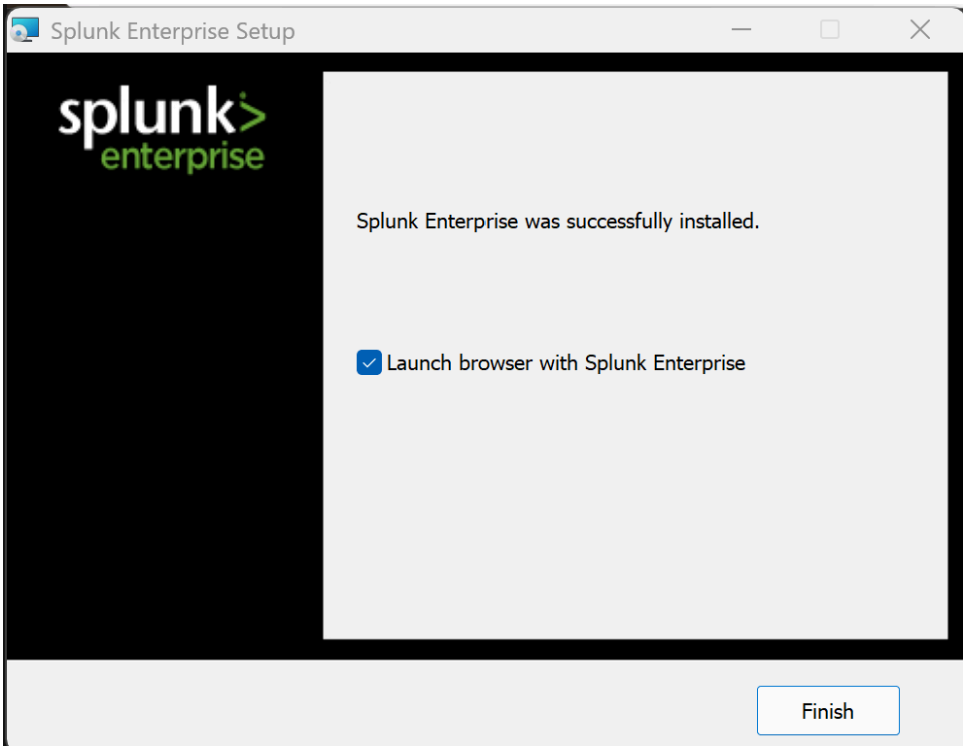
Step 1: Download and Set Up Splunk

- Visit the official Splunk website: https://www.splunk.com/en_us/download.html.
- Download and install the free trial version of Splunk Enterprise.

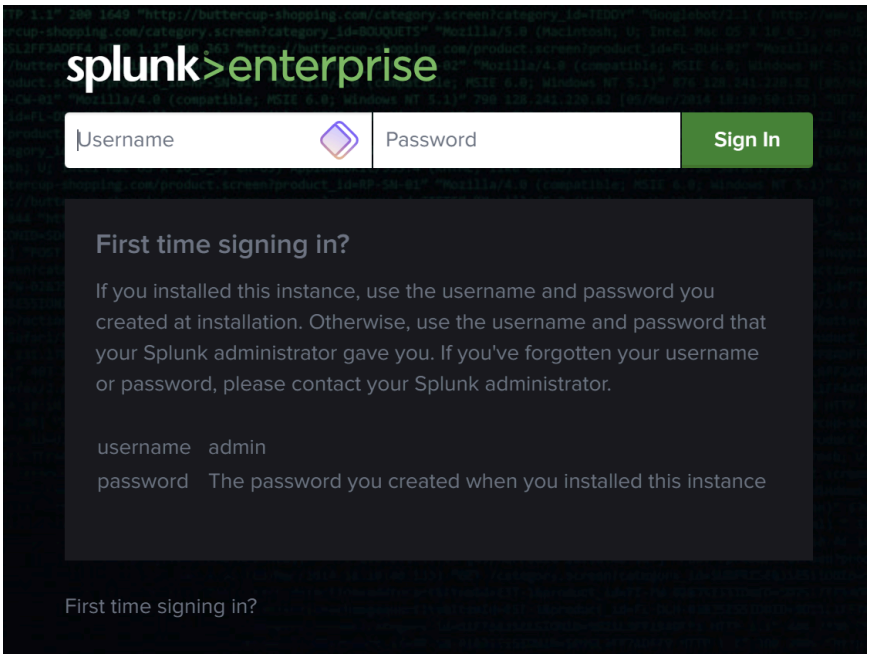


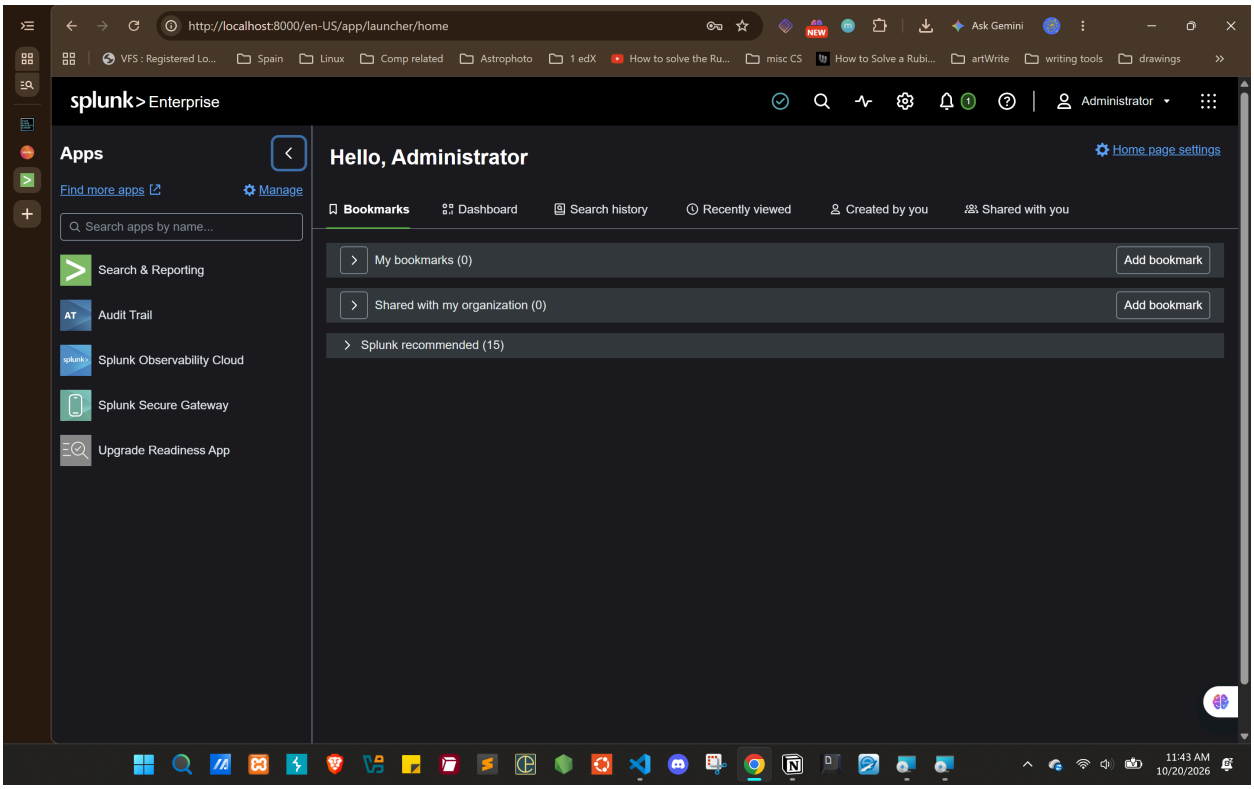
- You might need to register for an account on their website to download the free Splunk. When selecting the installation package, choose the one that suits your setup; in this case, I chose the Windows version.


https://www.splunk.com/en_us/download/splunk-enterprise.html
splunk>
Platform
Security
Observability
Industries
Resources
Pricing
Partners
Support
Trials & Downloads
GET STARTED
Choose Your Download
Splunk Enterprise 10.6.1.1
Index 500 MB/Day. Sign up and download now. After 60 days you can convert to a perpetual free license or purchase a Splunk Enterprise license to continue using the expanded functionality designed for enterprise-scale deployments.
Choose Your Installation Package
Windows
Linux
Mac OS
64-bit
Windows 10
Windows Server 2019, 2022, 2025
.msi
1102.78 MB
Download Now
Copy wget link
More
Release Notes
System Requirements
Previous Releases
All Other Downloads
Search
11:03 AM
10/20/2026


Splunk Enterprise Setup
splunk>
enterprise
Splunk Enterprise was successfully installed.
Launch browser with Splunk Enterprise
Finish

- After downloading, begin installing the downloaded file. You may be prompted to create an admin account before installation. Choose a username and password that are easy to remember.
- Open Splunk, which will direct you to the login page for your existing account. After you enter your username and password, you will be taken to the administrator dashboard.

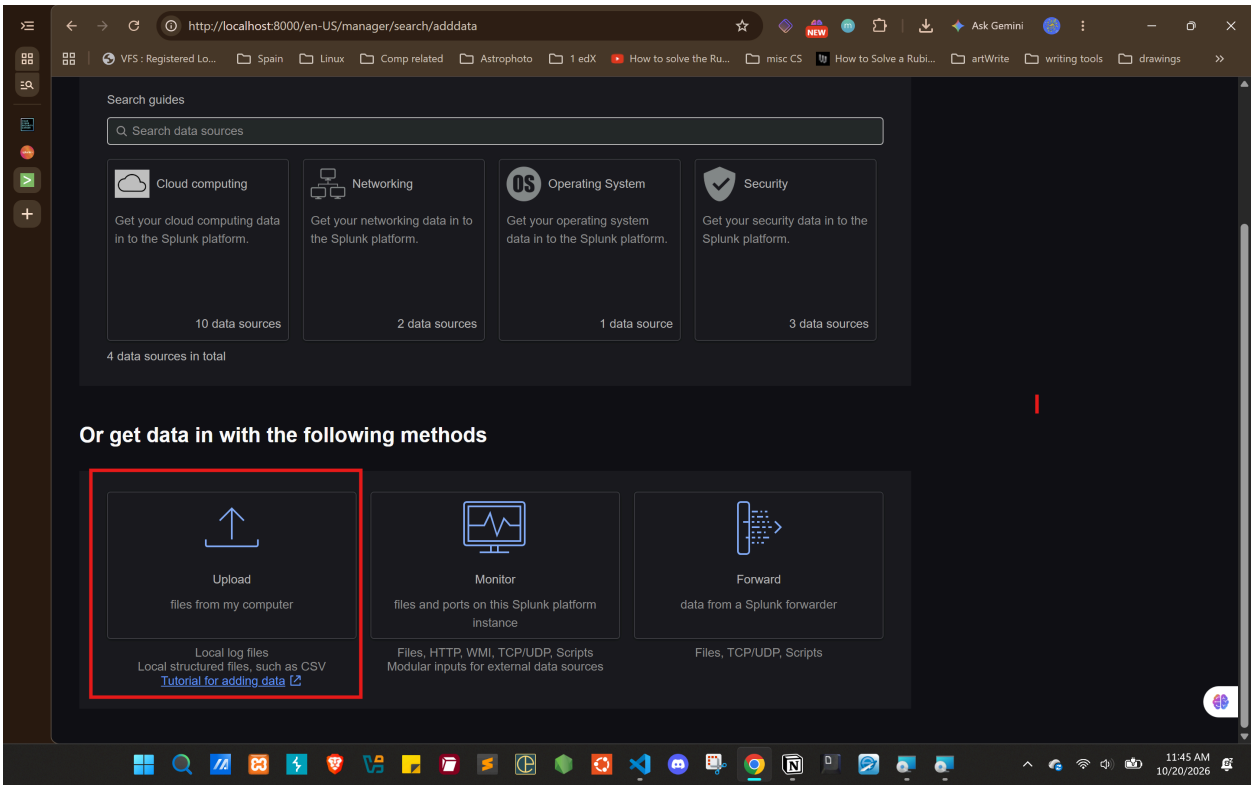


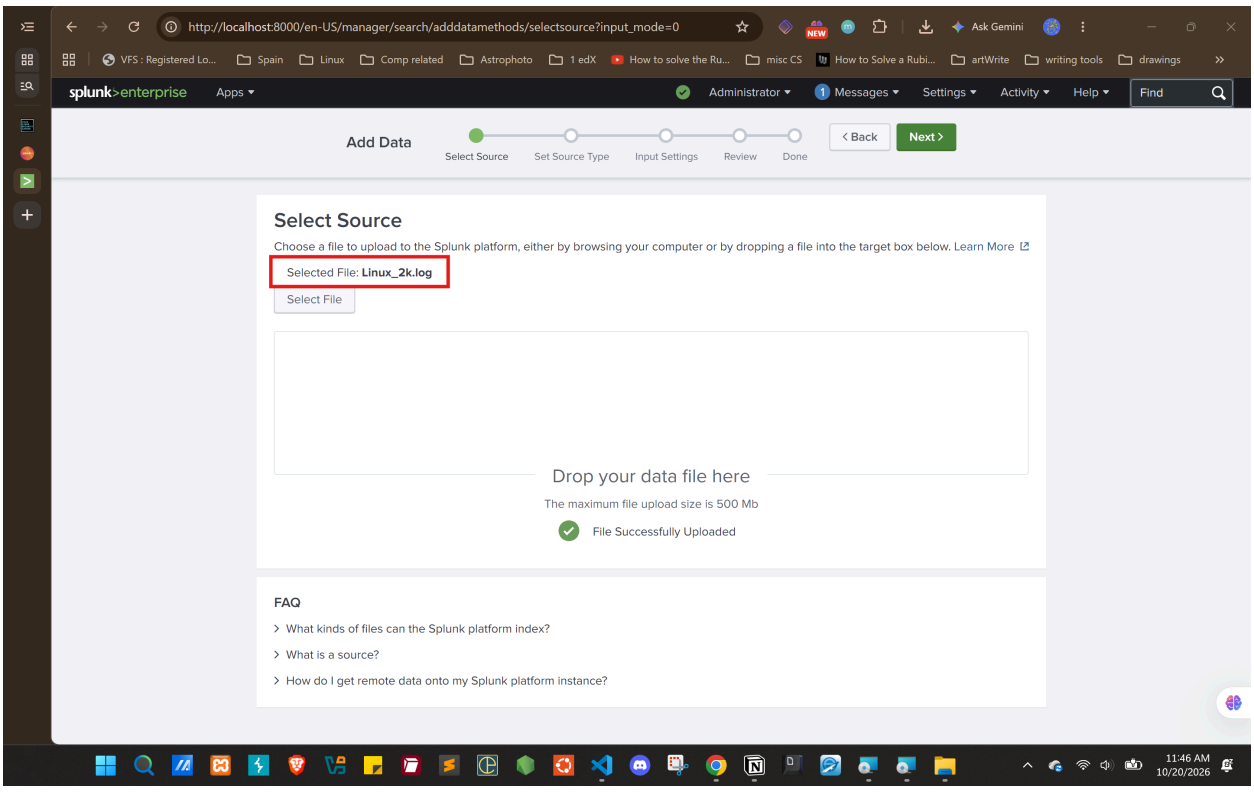


Step 2: Uploading the log file

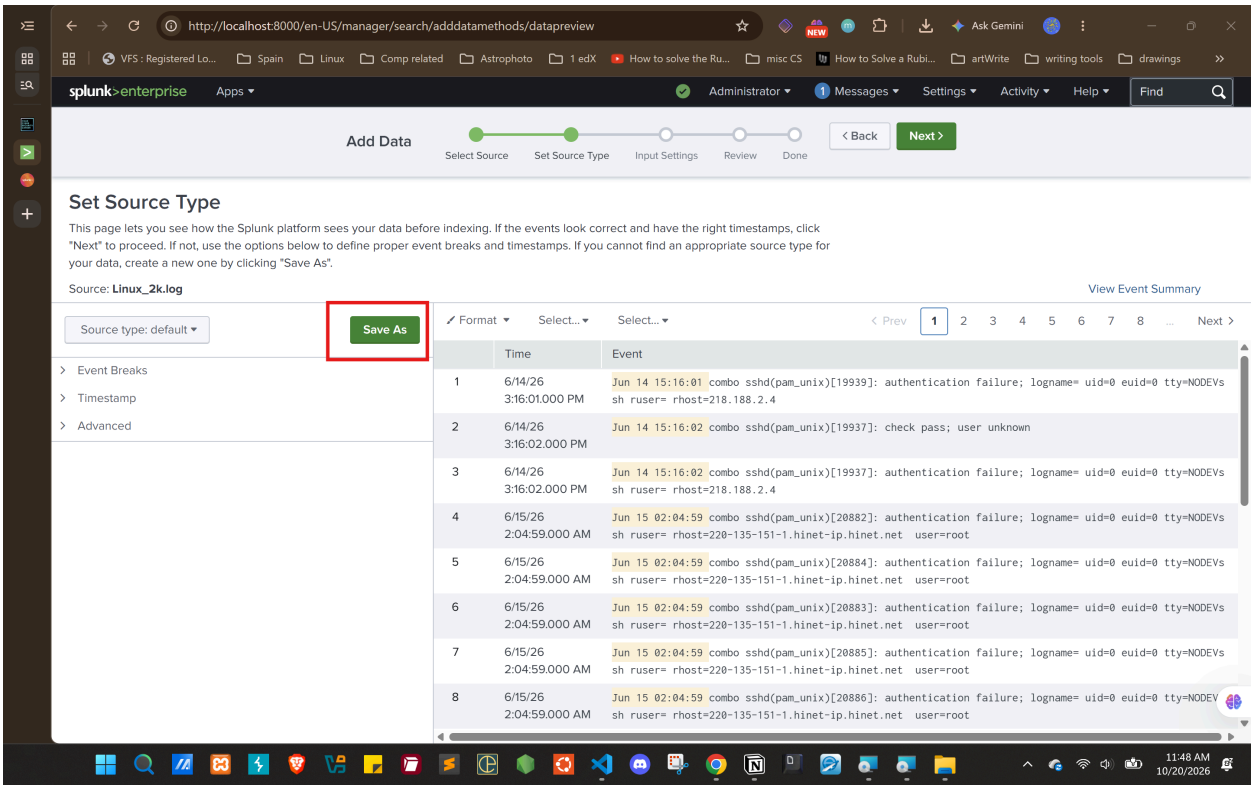
On the Splunk home page, go to **Settings → Add Data**

Select **Upload** and choose the same **Linux_2k.log** file used in the previous objectives, then proceed to **Source Type**.


http://localhost:8000/en-US/manager/search/adddata
Search guides
Search data sources
Cloud computing
Get your cloud computing data in to the Splunk platform.
10 data sources
Networking
Get your networking data in to the Splunk platform.
2 data sources
Operating System
Get your operating system data in to the Splunk platform.
1 data source
Security
Get your security data in to the Splunk platform.
3 data sources
4 data sources in total
Or get data in with the following methods
Upload
files from my computer
Local log files
Local structured files, such as CSV
Tutorial for adding data
Monitor
files and ports on this Splunk platform instance
Files, HTTP, WMI, TCP/UDP, Scripts
Modular inputs for external data sources
Forward
data from a Splunk forwarder
Files, TCP/UDP, Scripts


http://localhost:8000/en-US/manager/search/adddatamethods/selectsource?input_mode=0
splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Add Data
Select Source
Set Source Type
Input Settings
Review
Done
Back
Next
Select Source
Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. Learn More
Selected File: Linux_2k.log
Select File
Drop your data file here
The maximum file upload size is 500 Mb
File Successfully Uploaded
FAQ
What kinds of files can the Splunk platform index?
What is a source?
How do I get remote data onto my Splunk platform instance?

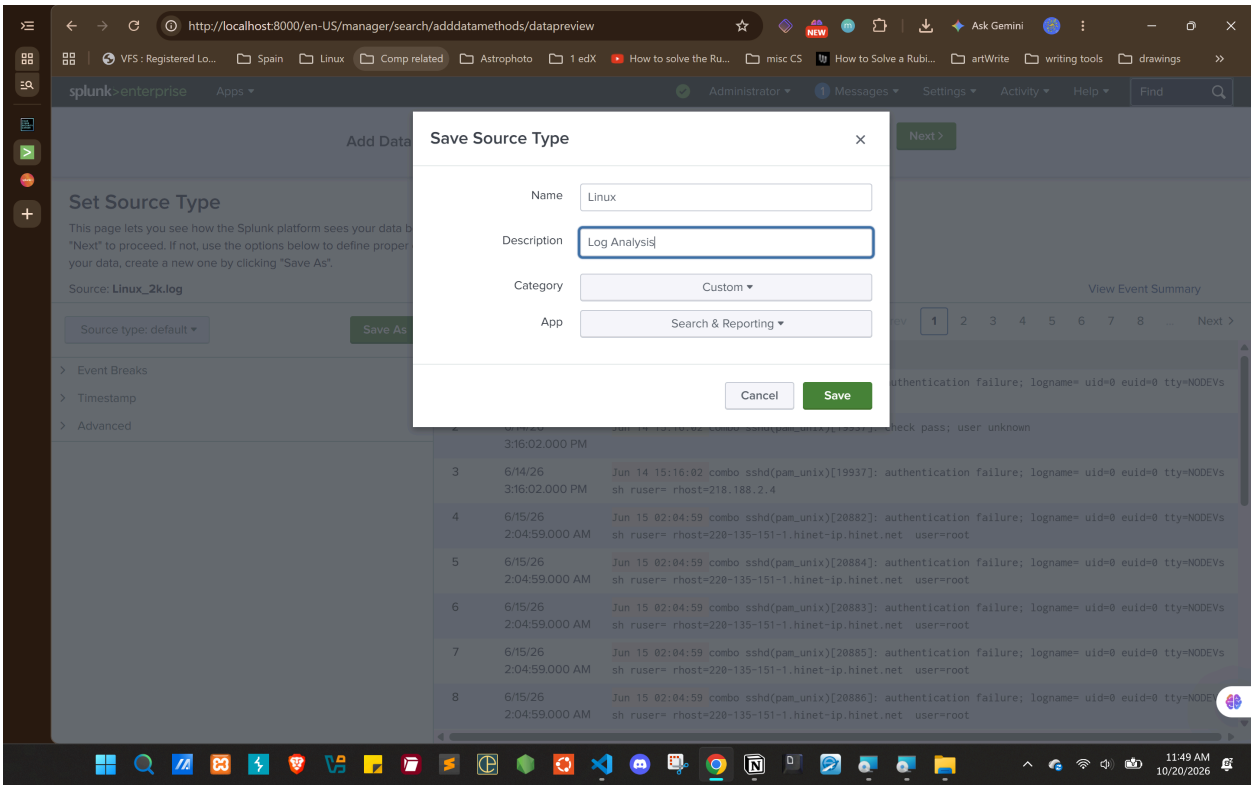
- Splunk will automatically detect and set the Source Type to its default value, then click Save As.

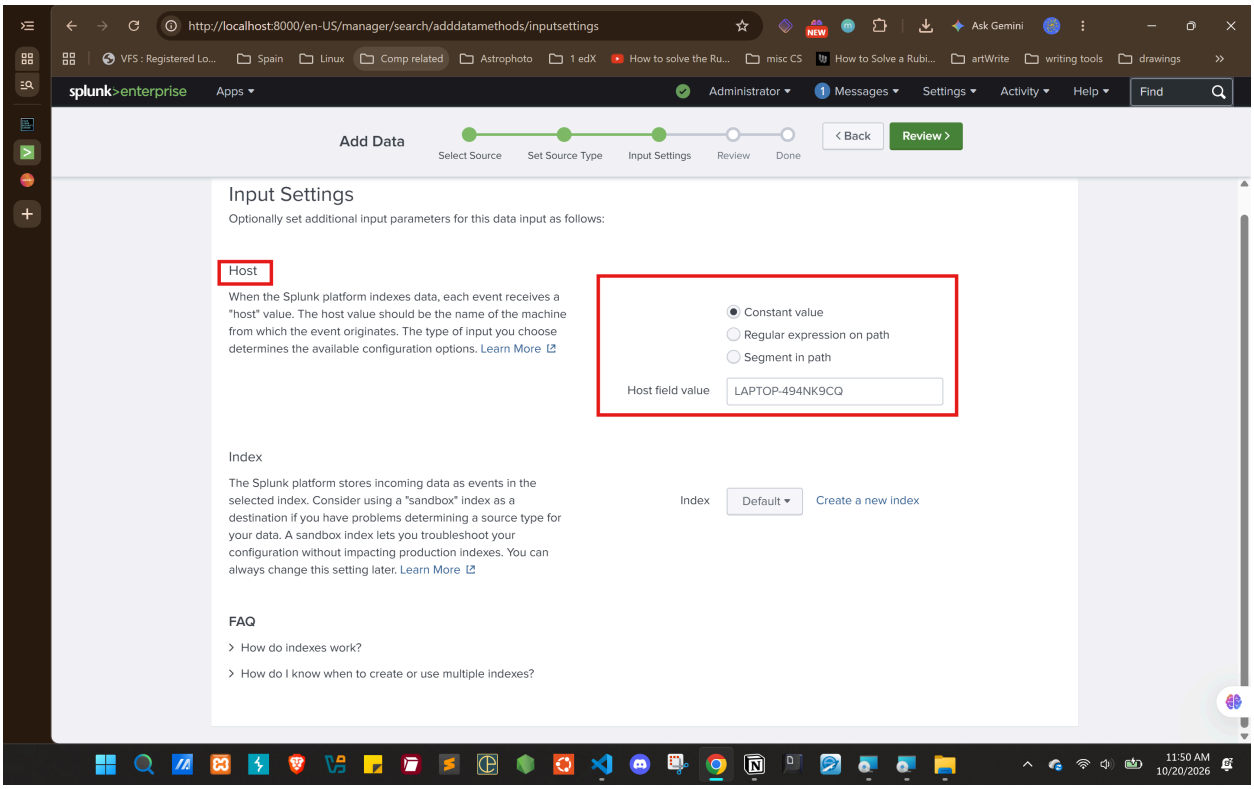


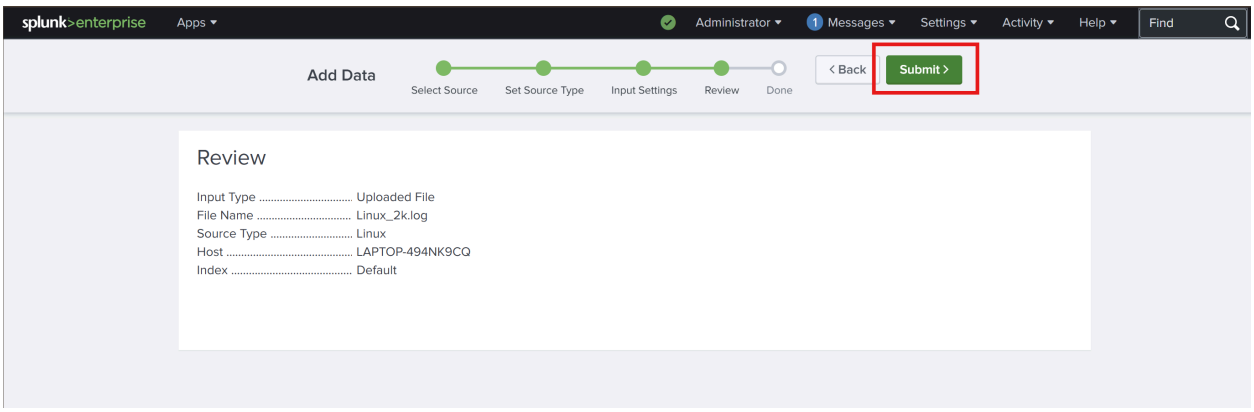
- Save this configuration with the following fields (as they appeared during setup):

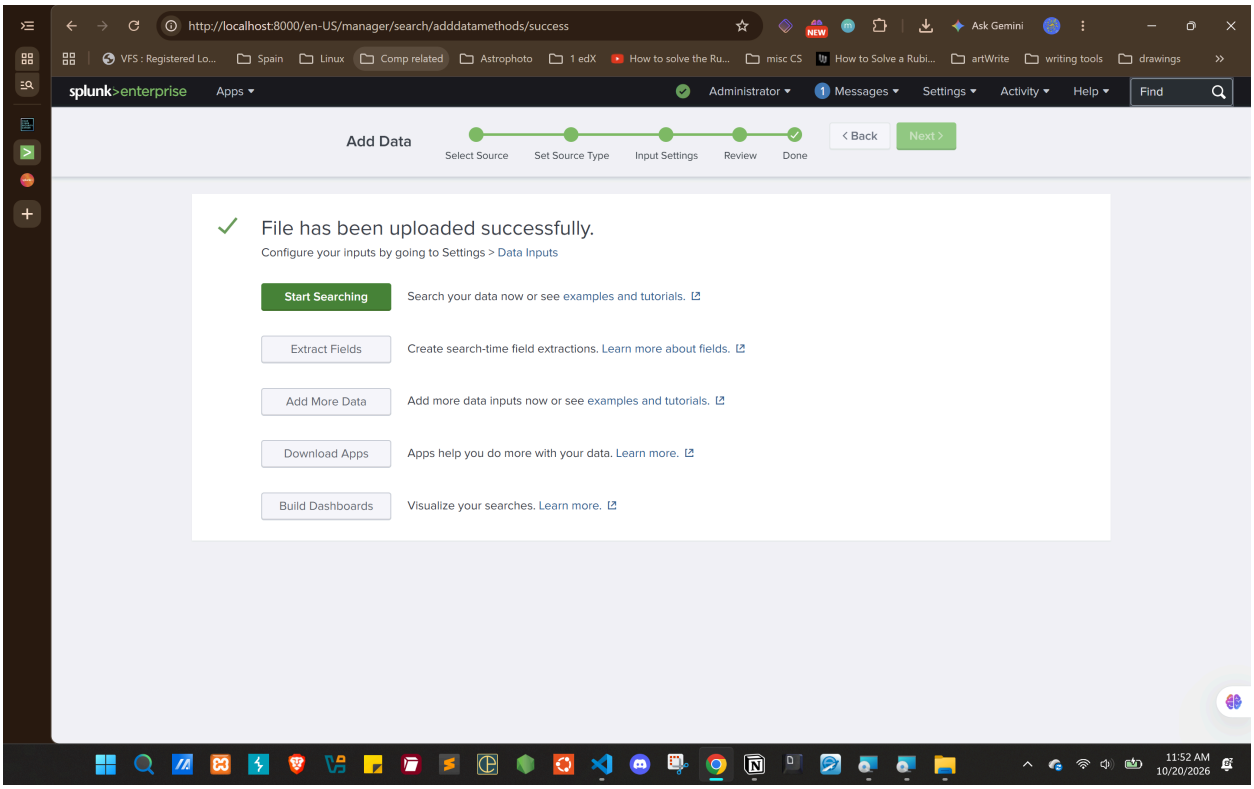
1. Source Type: (default auto-detected)
2. Name: Linux (or any name you choose)
3. Description: Log Analysis (describe it as you like)
4. Category: Custom
5. App: Search & Reporting
    - Continue to Input Settings, by setting the Host to a constant value (in my case it is LAPTOP-494NK9CQ - see below) and selecting the Default index. These settings ensure the log events are properly labeled and stored during ingestion. (Note: The constant host value will differ depending on the device you use.)

- Proceed to submit. If you encounter the error message "Upload failed with ERROR: can only concatenate str (not 'NoneType') to str," simply repeat the process and rename the log file to "Linux2k." Splunk may fail if the path or filename contains spaces, special characters, or non-standard symbols.



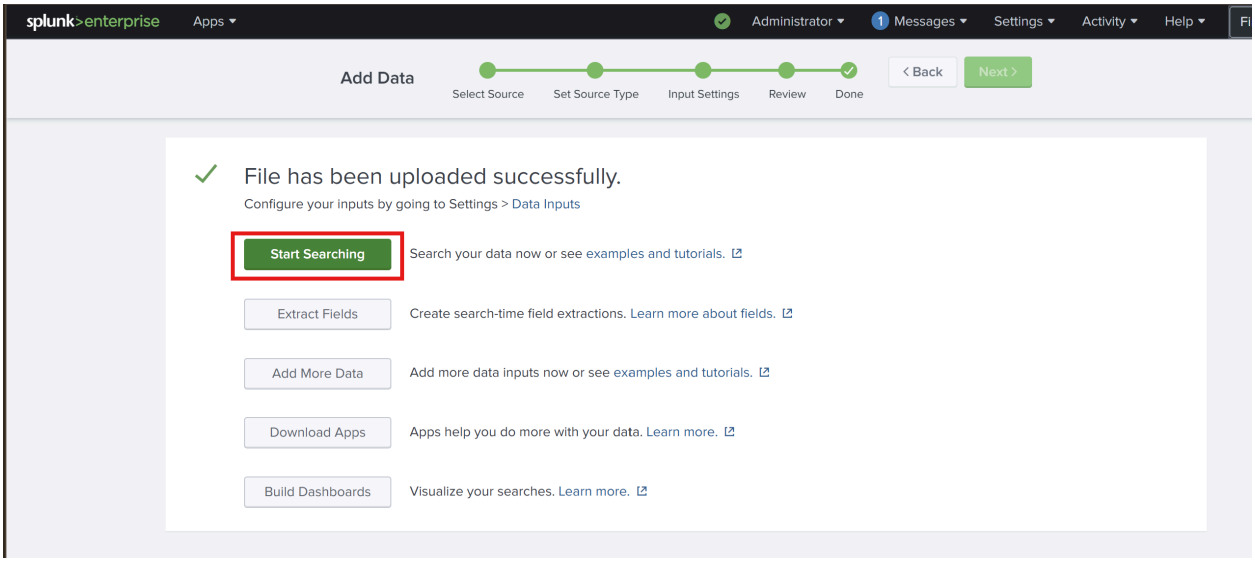

http://localhost:8000/en-US/manager/search/adddatamethods/inputsettings
splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Add Data
Select Source
Set Source Type
Input Settings
Review
Done
< Back
Review >
Input Settings
Optionally set additional input parameters for this data input as follows:
Host
When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. Learn More
Constant value
Regular expression on path
Segment in path
Host field value
LAPTOP-494NK9CQ
Index
The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. Learn More
Index
Default
Create a new index
FAQ
How do indexes work?
How do I know when to create or use multiple indexes?
11:50 AM
10/20/2026


splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Add Data
Select Source
Set Source Type
Input Settings
Review
Done
< Back
Submit >
Review
Input Type ..... Uploaded File
File Name ..... Linux_2k.log
Source Type ..... Linux
Host LAPTOP-494NK9CQ
Index ..... Default

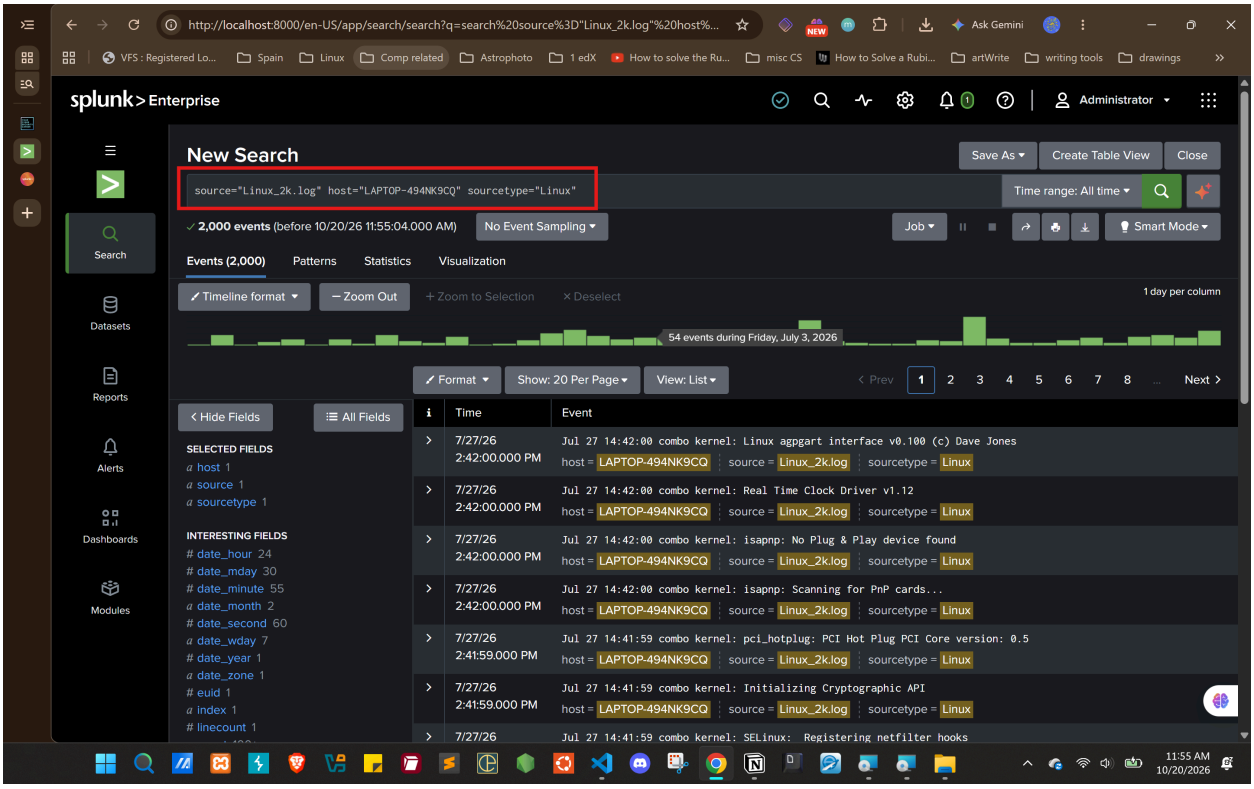


Step 3: Search and Filter Suspicious Activity

• When Splunk finishes indexing your data, your default search metadata will look like this:

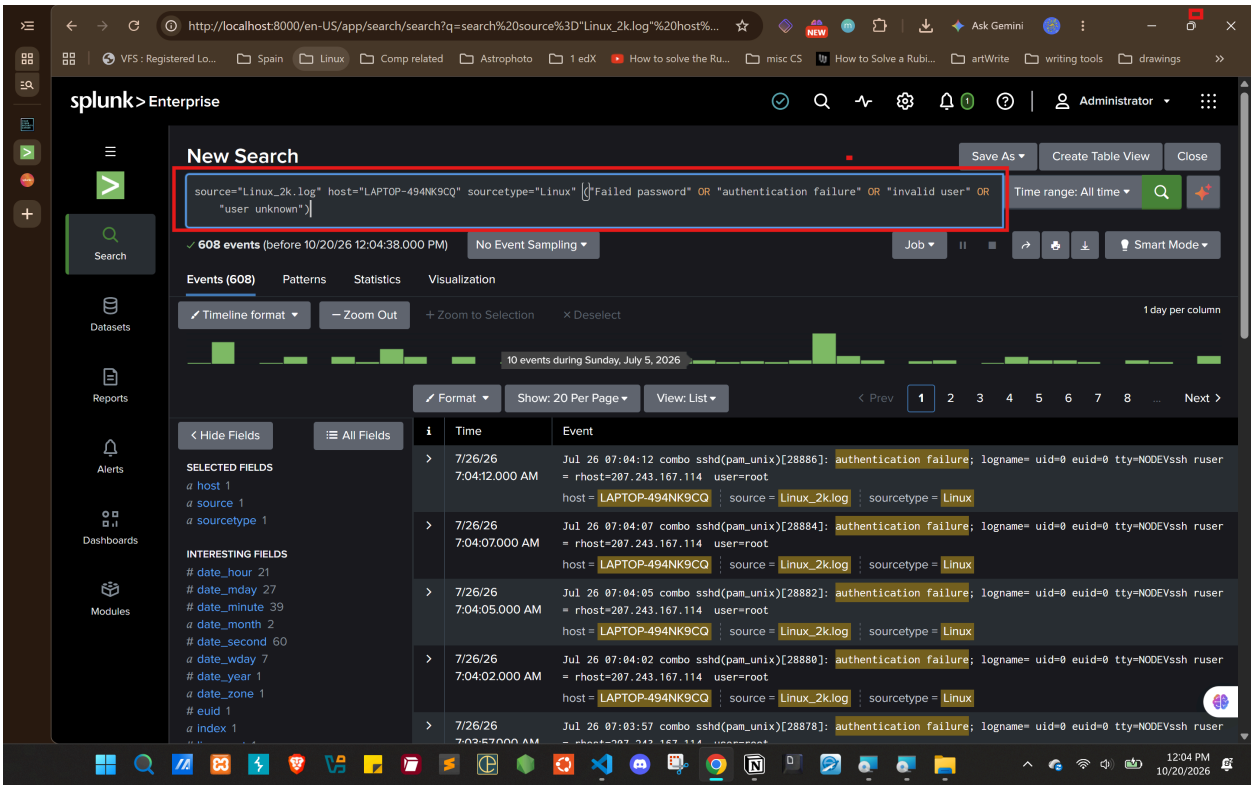


source="Linux_2k.log" host="LAPTOP-494NK9CQ" sourcetype="Linux”



• To search the log for suspicious authentication activity, use the following query in Splunk’s search bar:

source="Linux_2k.log" host="LAPTOP-494NK9CQ" sourcetype="Linux" ("Failed
password" OR "authentication failure" OR "invalid user" OR "user unknown")



This filters the log to show only entries that match common indicators of brute-force attempts or unauthorized access, such as:

- Repeated failed logins
- Invalid or unknown users
- Authentication failures

You can review these results to identify patterns in timestamps, usernames, and IP addresses.

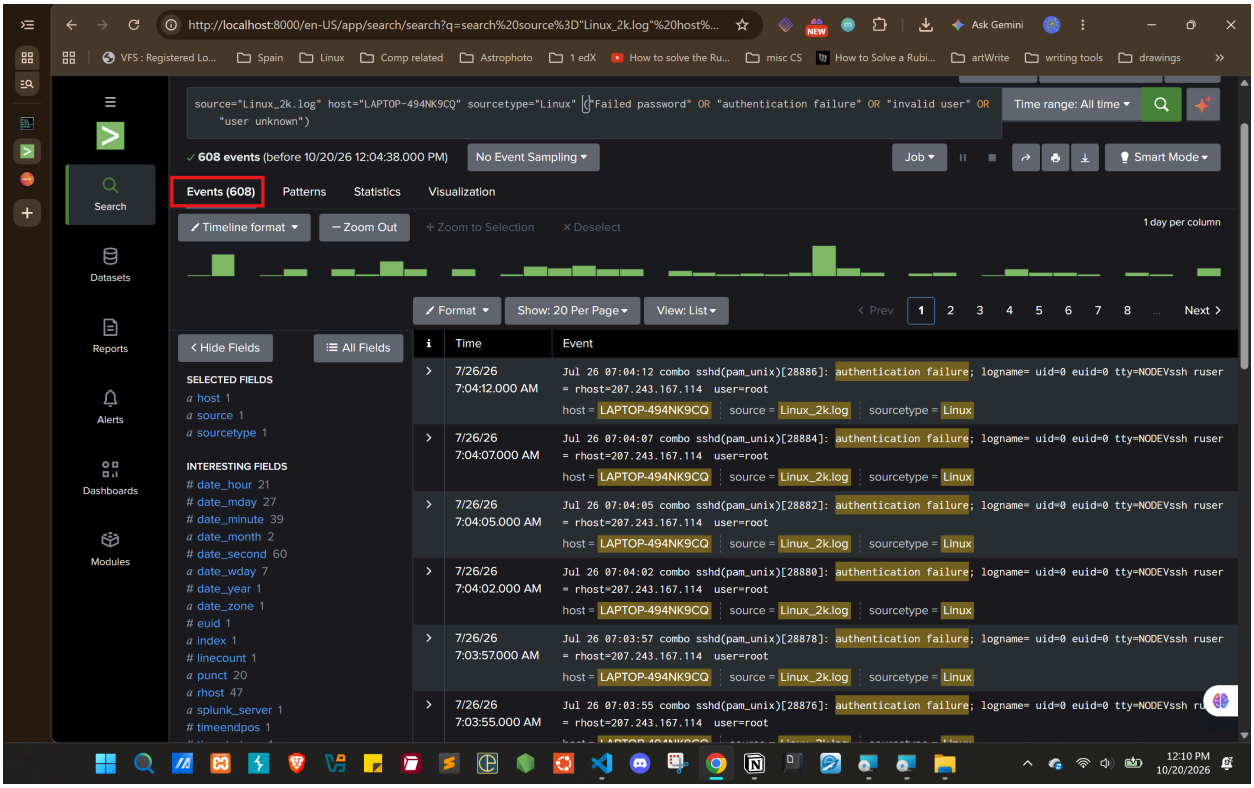
Step 4: Visualize the Data
After running the search query, Splunk shows all matching log entries under the Events tab.
One entry immediately stands out:

- Repeated failed login attempts
- Same username attempted: root

- Same source IP: 207.243.167.114
- Attempts occur within very short intervals (a few seconds apart)

This behavior is highly consistent with brute-force login attempts, so we follow this IP
deeper through the visualization tools.

Events View – Identifying the Suspicious Source

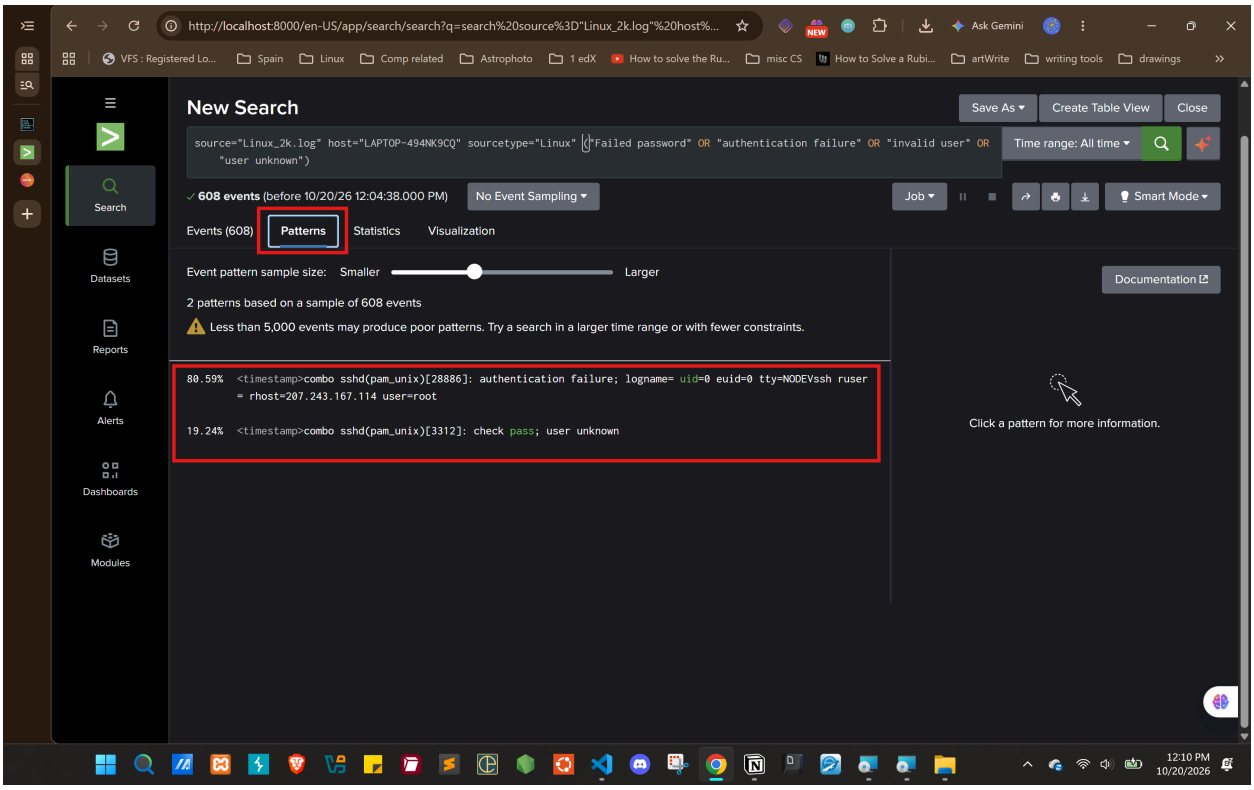


In the Events view, you can scroll through individual logs. Here, the IP 207.243.167.114 appears multiple times attempting to log into root. This raises a red flag because:

- “root” is a high-value target
- Repeated failures indicate likely unauthorized access attempts
  At this point, a SOC analyst would escalate this for pattern analysis.

Patterns View — Confirming Repetitive Behavior

Switching to Patterns, Splunk clusters similar events



One pattern appears prominently:

- "Failed password for root"

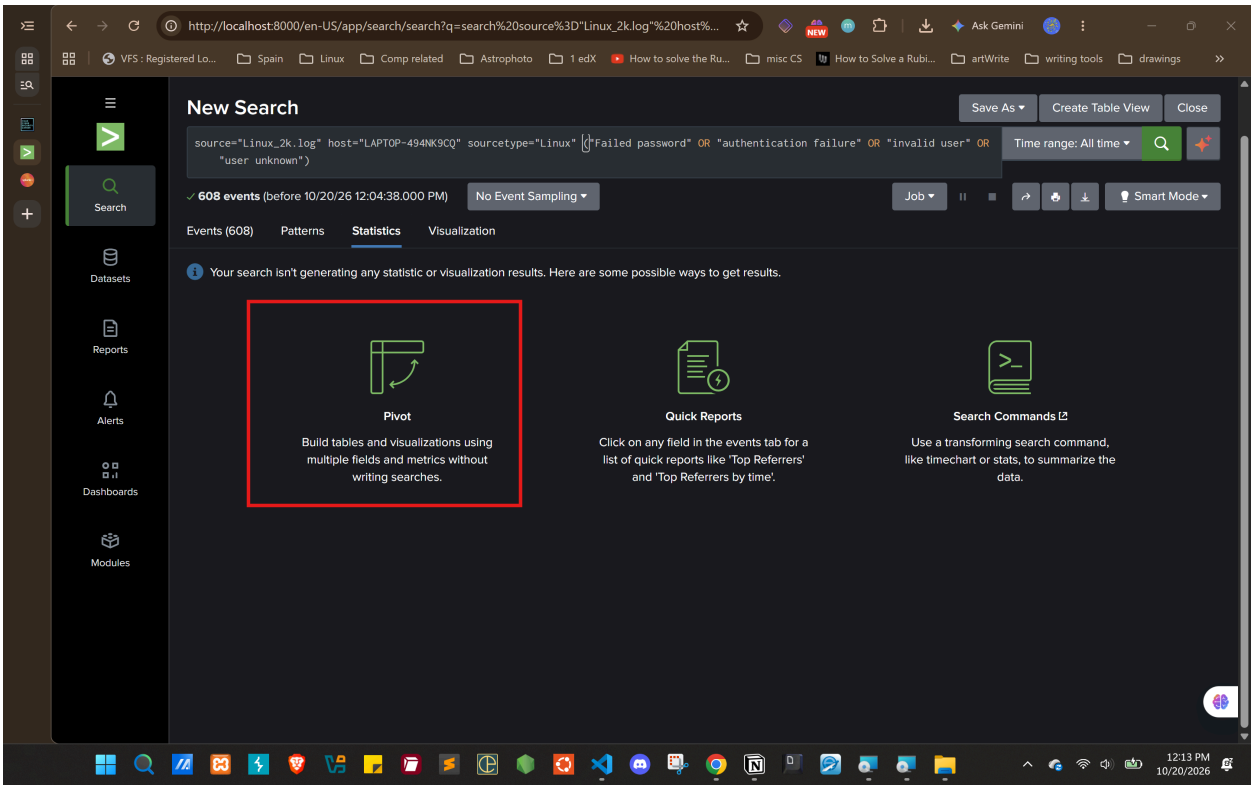
This confirms that multiple source IP generated repeated failures using the same target account. Seeing a repetition pattern allows analysts to determine the activity is not random — it's systematic.

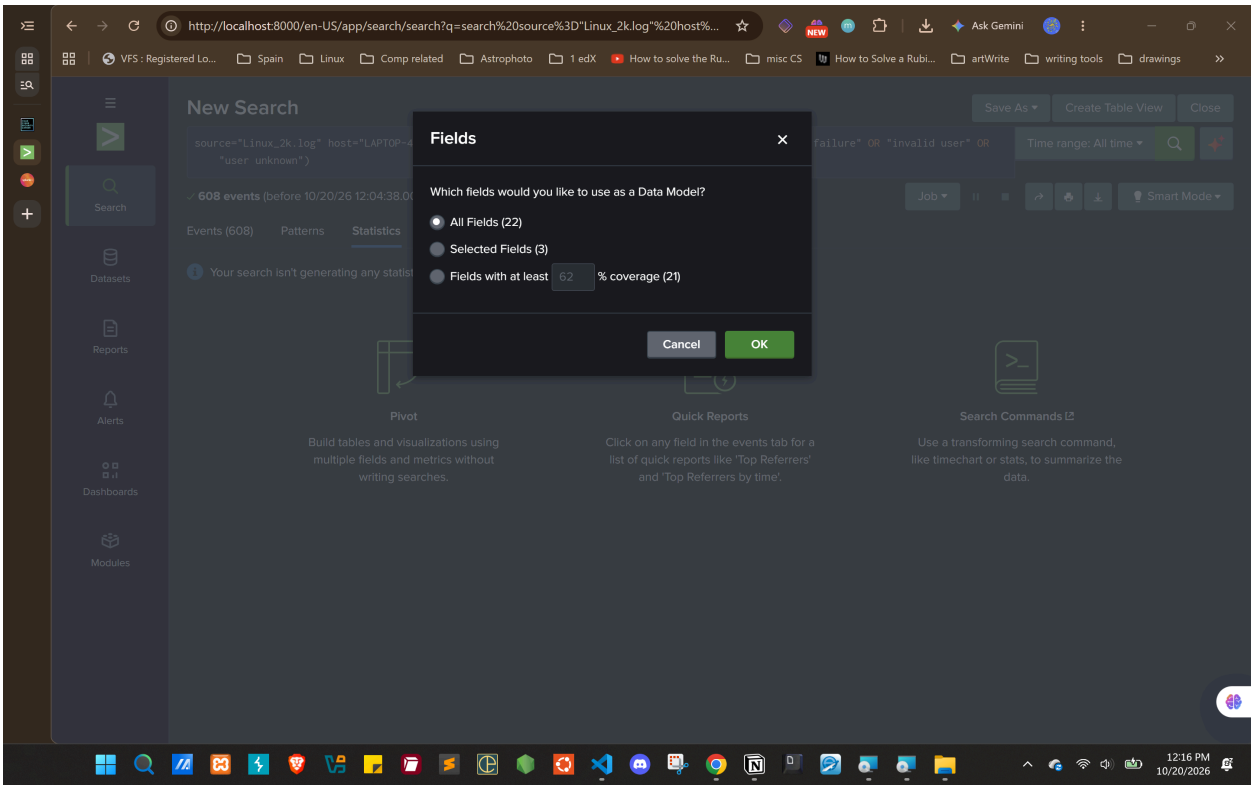
Statistics View – Breaking Down the Data

Switch to the Statistics tab to group events by IP, user, and event count. Observations include:

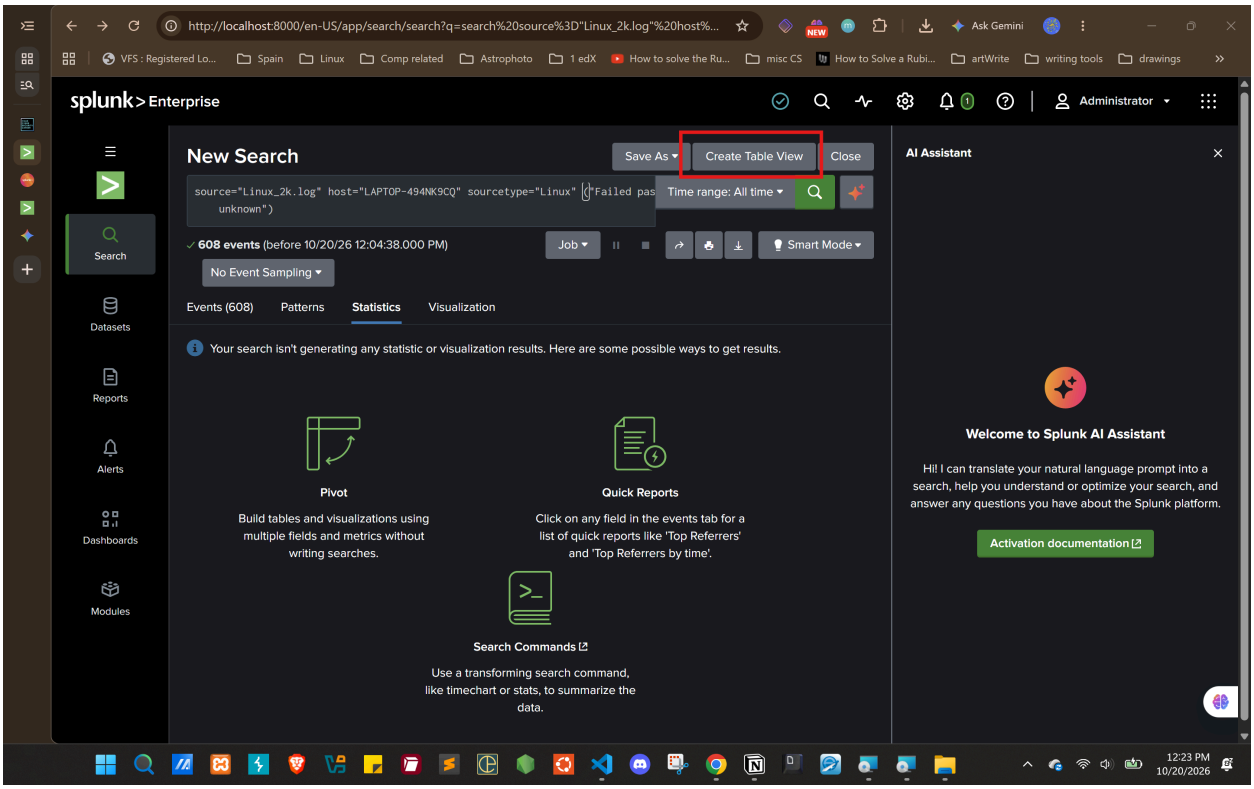
- 150.183.249.110 has the highest number of event count
- Targeted username is consistently root
- Other IPs show fewer and scattered attempts

This tabular view provides clarity on the scale and focus of the attack

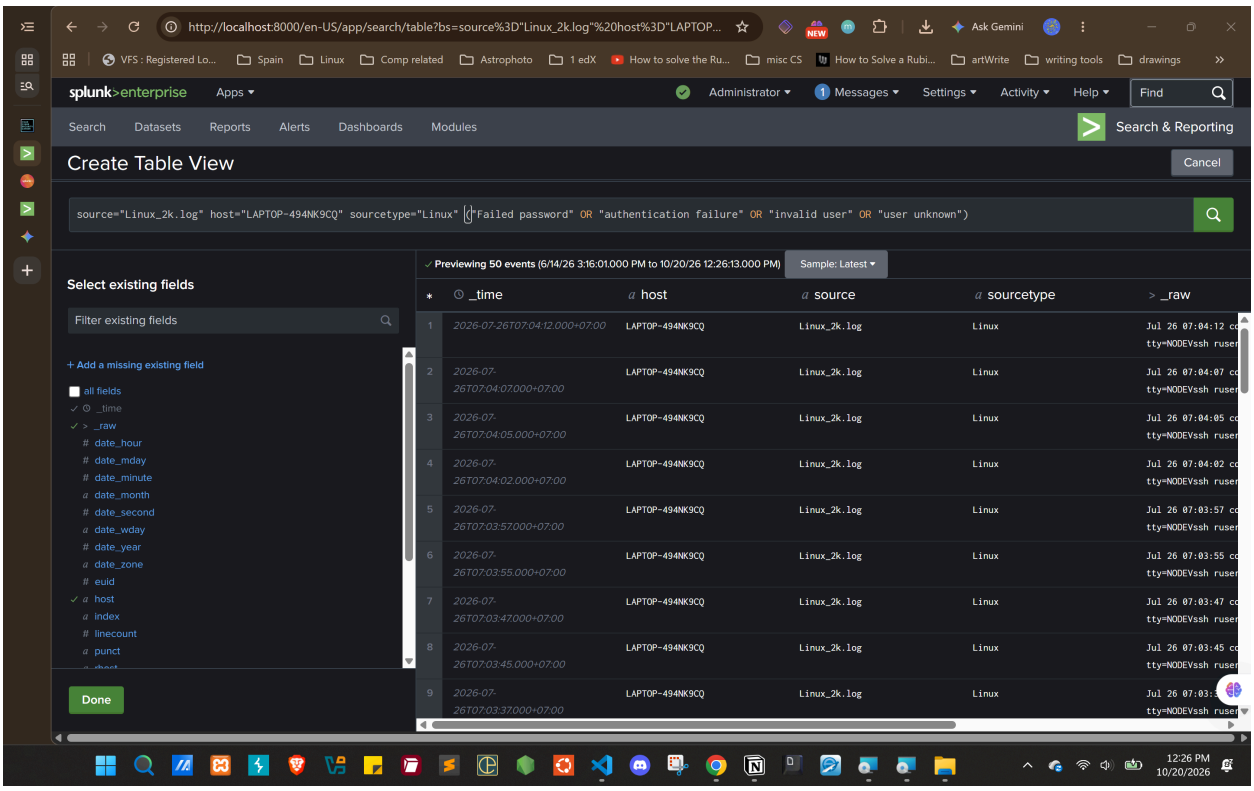




click on Create Table View

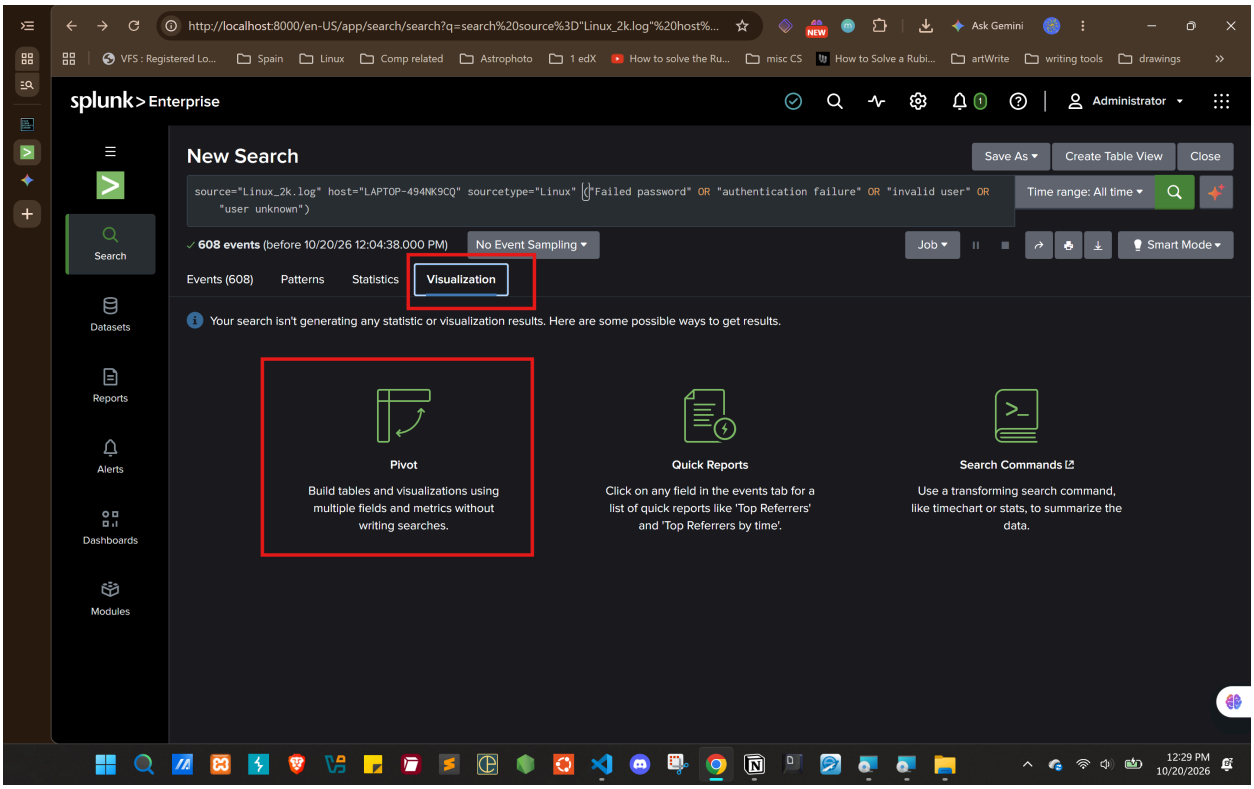


and you'll get this:

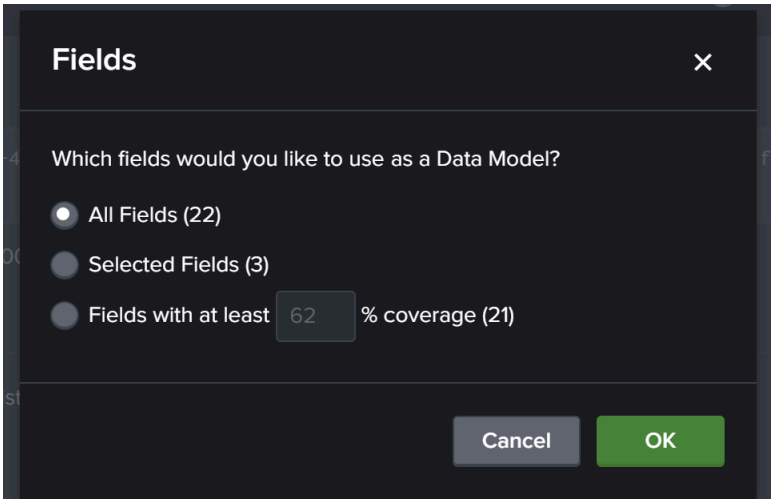


Visualization View – Presenting the Attack Timeline

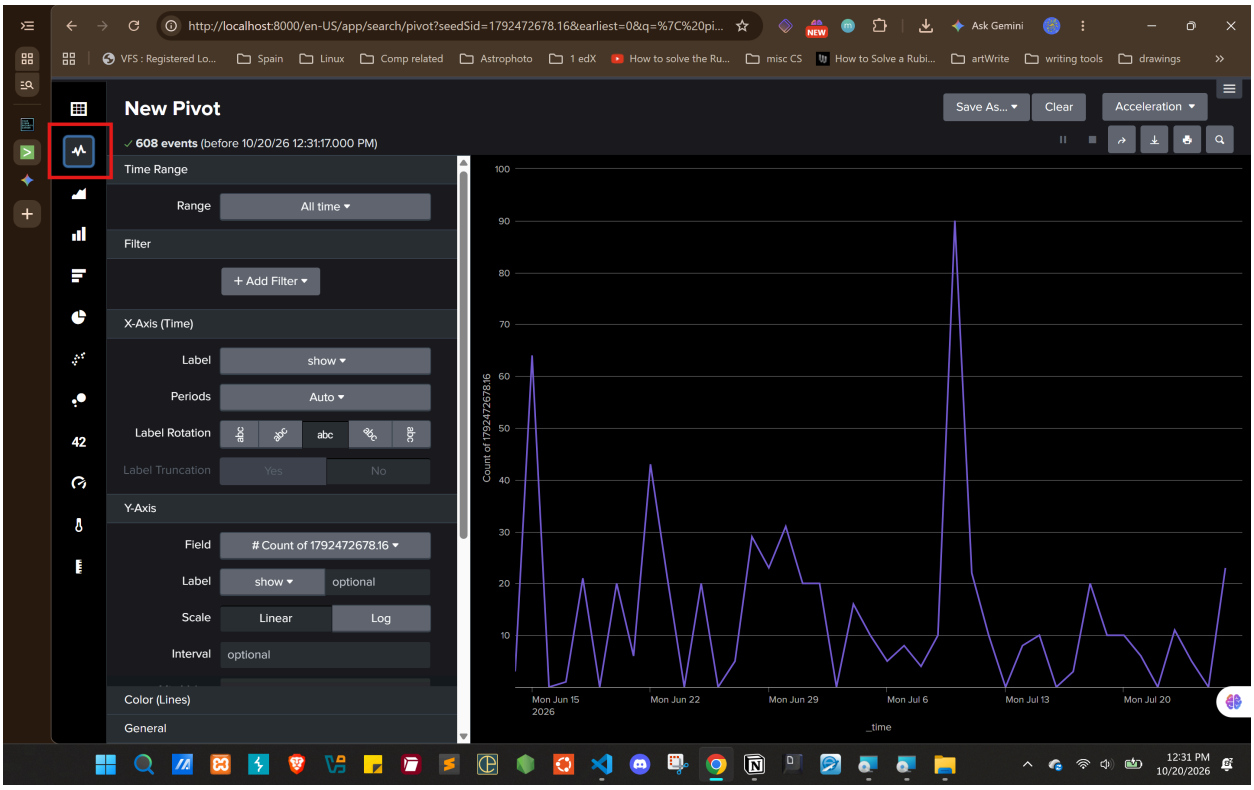
In the Visualization tab, a line chart is used to plot authentication failures over time. This visualization makes it easier to identify patterns and highlight potential security incidents



Select All Fields, OK:



click on the Line Chart icon and you'll get this:



Observations from the Line Chart:

- Sharp spikes in authentication failure events within short intervals

- Repeated attempts often target the root account
- These spikes correspond to multiple failed logins from the same IP address.

What This Suggests:

- The activity is time-based and bursty, indicating automated brute-force attempts
- The focus on the root username implies an attempt to gain administrative access
- This kind of trend is typical for attackers using scripts or botnets
  This form of visualization effectively demonstrates how SIEM tools can detect patterns at scale and help analysts prioritize threats by correlating activity over
  time.

Skills Learned

By completing this objective, you learned to:

- Ingest and index logs in Splunk Enterprise
- Use Splunk search syntax to filter authentication-related events
- Identify suspicious patterns using Events, Patterns, Statistics, and Visualization
- Generate visual insights for easier interpretation of attack patterns
- Integrate Splunk with manual and automated workflows from Objectives 1 and 2

Next Steps / Takeaways

- Review your findings from manual, automated, and Splunk-based log analysis to reinforce understanding of suspicious patterns.
- Experiment with larger log files or different Linux log datasets to practice scaling your analysis.

- Try creating more advanced Splunk searches, filters, or visualizations to simulate real SOC monitoring scenarios.
- Reflect on how manual, automated, and SIEM-based approaches complement each other in professional cybersecurity workflows